Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

2026-10-16. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*.

Question

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead, linear decay to 0 (conv only, or all four groups), one searched constant cooldown shared by all groups, and four searched constant cooldowns (one per group).

Setup

Item	Value
Model / data	CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules)
Intervals	4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead)
Candidates	7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline
Ground truth	Final test TTA when training continues on the baseline schedule after the interval
No lookahead	Test TTA right after the interval
Linear to 0	Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value
1 searched cooldown	Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed
4 searched cooldowns	As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes)
Seeds	4 training seeds (0-3); every method uses the same prefix and data order per seed
Score	Test-set TTA accuracy (%)

Results

Picked LR = the candidate with the highest seed-mean score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). Per-seed regret = the same when each seed picks from its own scores (what a single-seed search sees), averaged over seeds.

Picked LR (regret; per-seed regret)

Method	Interval 1	Interval 2	Interval 3	Interval 4
Ground truth (final TTA)	running	running	running	running
No lookahead	running	running	running	running
Linear to 0: conv only	running	running	running	running
Linear to 0: all four groups	running	running	running	running
1 searched cooldown (shared)	running	running	running	running
4 searched cooldowns (one per group)	running	running	running	running

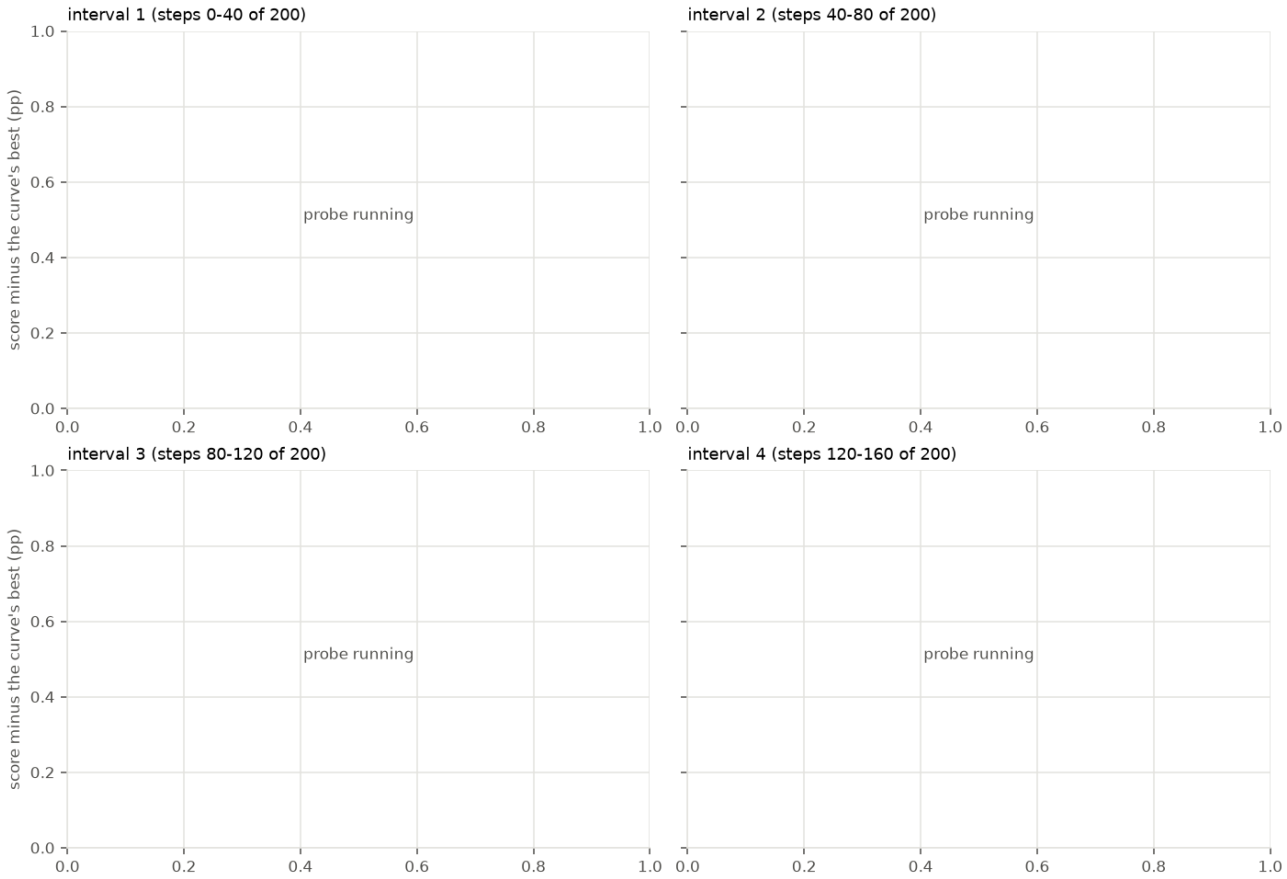


Figure 1. Each method's seed-mean score per candidate, relative to its own best (the point at 0 is its pick).